

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: FPGA Based System Design (01CT0718)	Aim: To design and implement digital code conversion systems for efficient data format transformation using FPGA.	
Experiment No: 06	Date:	Enrolment No: 92301733025

Aim: To design and implement digital code conversion system for efficient data format transformation using FPGA.

Software required:

Quartus sp13.0.1

Hardware required:

FPGA, USB-BLASTER

Theory:

A code converter is a combinational digital circuit that converts data from one binary code format to another without changing the actual information represented. Code converters are widely used in digital electronics, communication systems, microprocessors, and embedded systems where different devices use different coding standards. The aim of a code converter is to ensure compatibility and efficient data exchange between digital systems. The experiment is based on designing and implementing such a digital code conversion system on an FPGA using Verilog HDL.

FPGA (Field Programmable Gate Array) is an integrated circuit that can be programmed to perform various digital logic functions. By implementing code converters on an FPGA, the design becomes fast, flexible, reliable, and easy to modify without changing the hardware. The design is written in Verilog HDL, synthesized using Quartus II 13.0.1, simulated for functional verification, and finally programmed onto the FPGA board.

Some commonly used code converters are:

- **Binary to Gray Code Converter**
- **Gray to Binary Code Converter**
- **BCD to Excess-3 Converter**
- **Excess-3 to BCD Converter**
- **BCD to Seven-Segment Display Decoder**

Among these, the Binary-to-Gray and Gray-to-Binary converters are widely used because Gray code changes only one bit between consecutive numbers, reducing switching errors in applications such as rotary encoders, position sensors, and digital communication.

Binary to Gray Code Conversion

For an n-bit binary number:

- $G(n-1) = B(n-1)$ (Most Significant Bit remains the same)
- $G_i = B_{i+1} \oplus B_i$ (XOR operation)

Example:-

Binary	Gray
0000	0000
0001	0001
0010	0011
0011	0010
0100	0110

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Gray to Binary Code Conversion

To convert Gray code back to Binary:

- $B(n-1) = G(n-1)$
- $B_i = B_{i+1} \oplus G_i$

This method reconstructs the original binary number by performing cumulative XOR operations.

Advantages of FPGA-Based Code Converter

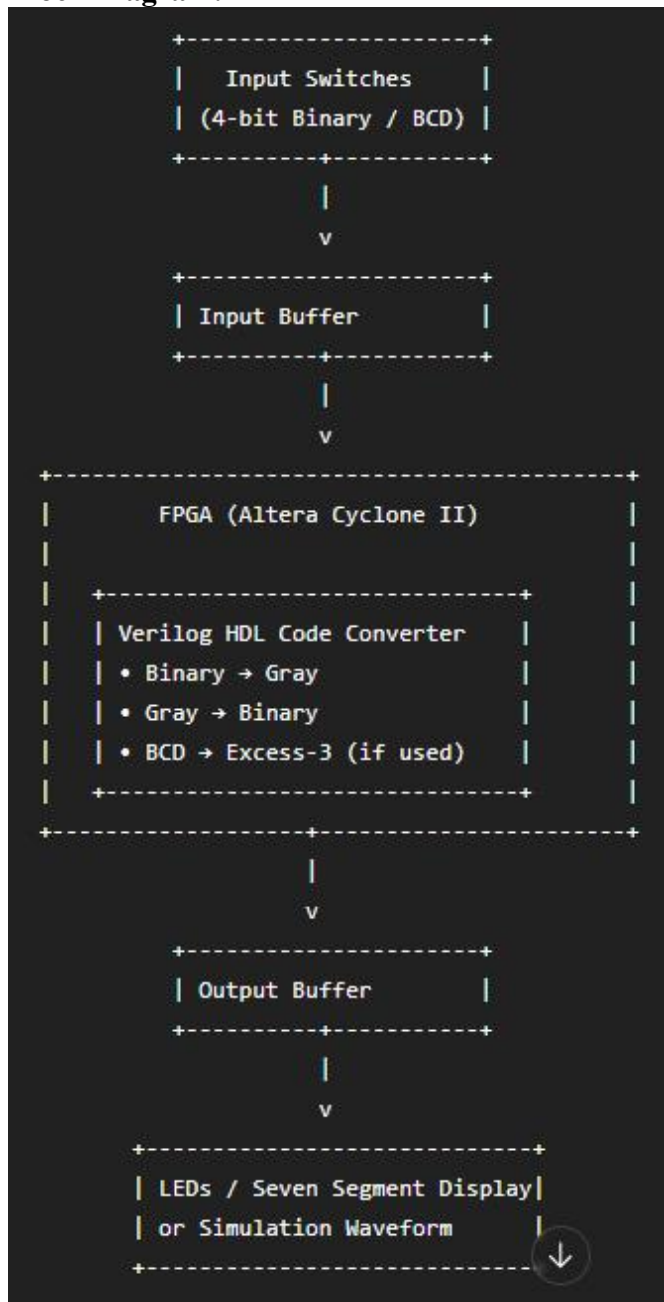
- High-speed data conversion
- Low hardware complexity
- Easy to modify and upgrade
- Reliable operation
- Suitable for real-time digital systems
- Efficient utilization of FPGA resources

Applications

- Digital communication systems
- Data transmission
- Error reduction in rotary encoders
- Microprocessor and microcontroller interfacing
- Digital signal processing (DSP)
- Embedded and automation systems

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Block Diagram:-



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Truth Table:-

Binary Input (B3 B2 B1 B0)	Gray Output (G3 G2 G1 G0)
0000	0000
0001	0001
0010	0011
0011	0010
0100	0110
0101	0111
0110	0101
0111	0100
1000	1100
1001	1101
1010	1111
1011	1110
1100	1010
1101	1011
1110	1001
1111	1000

Boolean Expression:-

$$G_3 = B_3$$

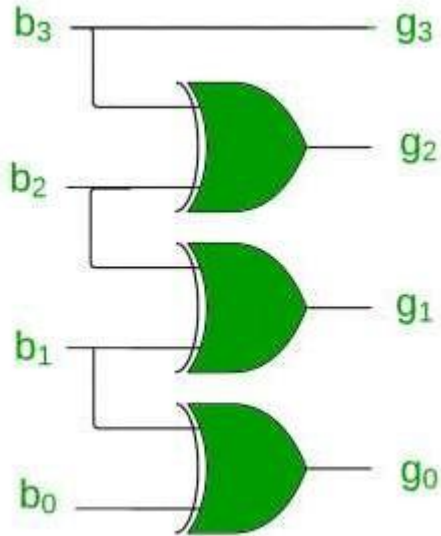
$$G_2 = B_3 \oplus B_2$$

$$G_1 = B_2 \oplus B_1$$

$$G_0 = B_1 \oplus B_0$$

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Circuit Diagram:-



Verilog Code:-

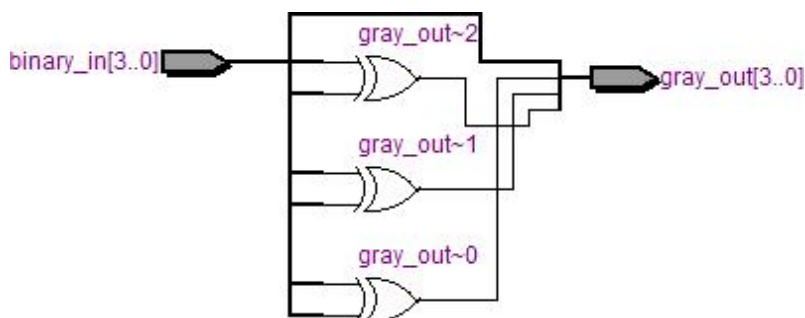
```

module Exp_6 (
    input [3:0] binary_in,
    output [3:0] gray_out
);
    assign gray_out[3] = binary_in[3];
    assign gray_out[2] = binary_in[3] ^ binary_in[2];
    assign gray_out[1] = binary_in[2] ^ binary_in[1];
    assign gray_out[0] = binary_in[1] ^ binary_in[0];
endmodule

```

Simulation Results:

(Simulation Waveform and RTL Schematic View)



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(FPGA Implementation with Pin Planning Details)

Conclusion:-

The Binary-to-Gray Code Converter was successfully designed and implemented on an FPGA using Verilog HDL. The simulation and hardware implementation verified correct code conversion for all inputs, demonstrating the FPGA's high speed, reliability, and suitability for real-time digital applications.

Post Lab Exercise:

1. Why are FPGA-based code converters preferred over discrete logic implementations in modern digital systems?

FPGA-based code converters are preferred because they provide high-speed operation, low hardware complexity, flexibility for modification, reliable performance, and efficient utilization of FPGA resources.

2. Why should invalid input combinations be explicitly handled in code converter designs?

Invalid input combinations should be handled to ensure reliable and error-free operation, preventing incorrect outputs and improving the stability of digital systems. *(This point is not explicitly discussed in the document; this answer is based on standard digital design practice.)*

3. Why are code converters used in digital systems?

Code converters are used to convert data from one binary code format to another without changing the actual information, ensuring compatibility and efficient data exchange between digital systems.

4. What is the advantage of gray code over binary code?

Gray code changes only one bit between consecutive numbers, reducing switching errors in applications such as rotary encoders, position sensors, and digital communication systems.

5. What is the purpose of assigning FPGA pins in Quartus?

FPGA pin assignment connects the Verilog input and output signals to the physical pins of the FPGA board, allowing the design to communicate with switches, LEDs, and other hardware components. The document also includes FPGA implementation with pin planning.

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6. How can the output of a code converter be verified on the FPGA board?

The output can be verified by applying binary inputs using the FPGA board switches and observing the Gray code outputs on LEDs or a seven-segment display. The design can also be verified through simulation waveforms in Quartus.